Chapter 7 supplementary material

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Three-Level Models

The two-level models described in Chapter 7 represented the association between *experience* and job-related *stress* among I = 1,000 nurses (Level 1) nested within J = 100 hospital wards (Level 2). But it turns out that the wards are nested in each of K = 25 hospitals (Level 3). Therefore, different wards are not independent Level 2 units because they are nested within hospitals.

Three-level intercept-only model

As explained Chapter 7, for two-level data, we can estimate an intercept-only model to partition outcome variable variance into Level 2 variance (variance between clusters) and Level 1 variance (variance across individuals within a cluster). Estimates of these variance parameters are used to calculate an *intraclass correlation* (ICC) summarizing the extent to which observations within a cluster are non-independent (i.e., proportion of between-cluster variance relative to total variance). The reduced-form equation for the two-level intercept-only model is:

$$Y_{ij} = \gamma + \delta_j + \varepsilon_{ij}$$

The total variance of Y can be partitioned into two variance components:
$\mathrm{VAR}(Y_{ij}) = \mathrm{VAR}(\gamma + \delta_j + \varepsilon_{ij}) = \mathrm{VAR}(\delta_j) + \mathrm{VAR}(\varepsilon_{ij}) = \tau^2 + \sigma^2$
The ICC represents the proportion of total variance due to clustering as
$ICC = \frac{\tau^2}{\tau^2+\sigma^2}$.

The **three-level intercept-only model** can be represented with the following equations:
Level 1: $Y_{ijk} = \beta_{0jk} + \varepsilon_{ijk}$
where Y_{ijk} is the value of the outcome variable for individual i in cluster j within macro-unit k (e.g., *stress* for nurse i in ward j in hospital k), β_{0jk} is the mean (i.e., random intercept) in cluster j within macro-unit k (e.g., mean *stress* in ward j), and ε_{ij} is the individual-level error (i.e., difference between an individual's observed value and β_{0jk}).
Level 2: $\beta_{0jk} = \gamma_{00k} + \delta_{0jk}$
where γ_{00k} is the mean in macro-unit k and δ_{0jk} is the Level-2 (e.g., ward-level) random effect, that is, the difference between the mean in cluster j and the mean in macro-unit k (e.g., difference between the mean in ward j and the mean of the corresponding hospital).
Level 3: $\gamma_{00k} = \mu_{000} + \varsigma_{00k}$
where μ_{000} is the grand mean of the outcome variable and ζ_{00k} is the Level-3 (e.g., hospital-level) random effect, that is, the difference between the mean in hospital k and the grand mean.
Reduced form (i.e., combined equation): $Y_{ijk} = \mu_{000} + \zeta_{00k} + \delta_{0jk} + \varepsilon_{ijk}$
Now the total variance is partitioned into *three* variance components:

$\mathrm{VAR}(Y_{ijk}) = \mathrm{VAR}(\mu_{000} + \zeta_{00k} + \delta_{0jk} + \varepsilon_{ijk}) =$
$\mathrm{VAR}(\zeta_{00k}) + \mathrm{VAR}(\delta_{0jk}) + \mathrm{VAR}(\varepsilon_{ijk}) = \tau_\zeta^2 + \tau_\delta^2 + \sigma^2$.
In words, total variance equals the sum of Level 3 variance (τ_ζ^2), Level 2 variance (τ_δ^2), and Level 1 variance (σ^2).

Next, the intraclass correlation concept is necessarily more complex with three variance components instead of just two. With two-level data, the ICC represents the extent to which Level 1 observations are non-independent due to nesting within Level 2 clusters. In this three-level situation, we can summarize the extent to which Level 1 observations are non-independent due to the overall nesting within *both* Level 2 (wards) and Level 3 units (hospitals) by calculating the proportion of variance due to the sum of Level 2 and Level 3 variance:

$$\widehat{ICC}_{L2+L3} = \frac{\hat{\tau}_\zeta^2 + \hat{\tau}_\delta^2}{\hat{\tau}_\zeta^2 + \hat{\tau}_\delta^2 + \hat{\sigma}_\varepsilon^2}$$

For the nurse example, 69% of total variance in stress levels is due to the sum of between-ward and between-hospital variance.

Additionally, we might be interested in separate estimates of the proportion of variance due to Level 2 nesting and the proportion of variance due to Level 3 nesting:

$$\widehat{ICC}_{L2} = \frac{\hat{\tau}_\delta^2}{\hat{\tau}_\zeta^2 + \hat{\tau}_\delta^2 + \hat{\sigma}_\varepsilon^2}$$

and

$$\widehat{ICC}_{L3} = \frac{\hat{\tau}_\zeta^2}{\hat{\tau}_\zeta^2 + \hat{\tau}_\delta^2 + \hat{\sigma}_\varepsilon^2}.$$

For the nurse example, $ICC_{L2} = 0.51$ is the proportion of variance due to nesting of nurses within wards, and $ICC_{L3} = 0.18$ is proportion of variance due to nesting of wards within hospitals.

Conditional three-level models

Because of the considerable variance between hospitals (i.e., different wards are not independent Level 2 units because they are nested within hospitals; $ICC_{L3} = 0.18$), we should revise the random-slopes two-level model presented in Chapter 7 to represent the association between *experience* and *stress* within a three-level model:

Level 1: $Y_{ijk} = \beta_{0jk} + \beta_{1jk} X_{ijk\bullet} + \varepsilon_{ijk}$

$stress_{ijk} = \beta_{0jk} + \beta_{1jk} experience_{ijk\bullet} + \varepsilon_{ijk}$

Level 2a: $\beta_{0jk} = \gamma_{00k} + \delta_{0jk}$

Level 2b: $\beta_{1jk} = \gamma_{10k} + \delta_{1jk}$

Level 3a: $\gamma_{00k} = \mu_{000} + \varsigma_{00k}$

Level 3b: $\gamma_{10k} = \mu_{100} + \varsigma_{10k}$

Now, we see that the intercept β_{0jk} in the Level 1 equation can vary across both Level 2 units (wards), indexed by j, and Level 3 units (hospitals), indexed by k. Specifically, β_{0jk} is the intercept of ward j within hospital k; γ_{00k} is the mean intercept of the wards in hospital k; and μ_{000} is the grand mean intercept (i.e., the fixed intercept). Likewise, the slope β_{1jk} in the Level 1 equation can vary across both Level 2 units and Level 3 units. Specifically, β_{1jk} is the *experience* slope for ward j within hospital k; γ_{10k} is the mean slope of the wards in hospital k; and μ_{100} is the grand mean slope (i.e., the fixed slope).

Each of the Level 2 and Level 3 equations includes an error term, indicating that the within-ward intercepts vary from their respective mean intercept (Level 2a equation) *and* the within-hospital intercepts vary from the grand mean intercept (Level 3a equation); consequently, we have random-intercept variance parameters for both Level 2 and Level 3. Likewise, the within-ward slopes vary around their respective ward-specific mean slope (Level 2b equation) *and* the within-hospital slopes vary around the grand mean mean slope (Level 3b equation); consequently, we have random-slope variance parameters for both Level 2 and Level 3. Furthermore, the full model will include a covariance between the within-ward intercepts and slopes and a covariance between the within-hospital intercepts and slopes. Yet, the data may not support inclusion of all of these variance parameters, either with respect to model convergence or with respect to the statistical significance of the random effects.

When this model is fitted to the nurse data (with experience again centered within-wards), the grand mean slope = -0.0237 is very close to the fixed slope estimate we obtained for the two-level model (-0.0236), but the standard error of this estimate is now slightly higher (0.003194 vs. 0.003085) because the non-independence of wards within hospitals is now taken into account. But the between-ward slope standard deviation = 0.008 is quite small, even smaller than it was from the two-level model, and the between-hospital slope standard deviation = 0.005 is even smaller than that. In fact, the model solution includes a perfect correlation between hospital-level intercepts and slopes, suggesting that having Level 3 random slopes is not well supported by the data.

In general, it's considered best practice to estimate the "maximal" random-effects structure that's justified by the study design (see Barr et al., 2013). In the nurse example, the maximal random-effects structure includes both Level 2 and Level 3 intercept and slope variance. But it commonly occurs that the model with maximal random effects is too complex to produce a converged solution with proper parameter estimates; in the nurse example, the perfect correlation between Level 3 intercepts and slopes is improper. Consequently, a simpler, but only minimally simpler, model should be estimated. Often, it suffices to restrict the slope variance at the highest level to 0.

Thus, perhaps we should simplify the three-level model nurse to include only random intercepts at Level 3, fixing the Level 3 slope to be constant across hospitals:

Level 1: $Y_{ijk} = \beta_{0jk} + \beta_{1jk}X_{ijk\bullet} + \varepsilon_{ijk}$

$stress_{ijk} = \beta_{0jk} + \beta_{1jk}experience_{ijk\bullet} + \varepsilon_{ijk}$

Level 2a: $\beta_{0jk} = \gamma_{00k} + \delta_{0jk}$

Level 2b: $\beta_{1jk} = \gamma_{10k} + \delta_{1jk}$

Level 3a: $\gamma_{00k} = \mu_{000} + \varsigma_{00k}$

Level 3b: $\gamma_{10k} = \mu_{100}$

These equations are identical to those given on the previous page, except the Level 3b equation no longer includes an error term, indicating that the experience slope is considered constant across hospitals. That is, the experience effect still varies across wards, but only intercepts vary across both wards and hospitals.

Fitting this revised three-level model to the nurse data results in a proper, converged solution. The grand mean slope estimate remains -0.0237, whereas its standard error has changed very slightly. Removing the between-hospital slope variance has led to a slight increase in the between-ward slope standard deviation (now 0.009 vs. 0.008 in the previous model).

Higher-level random slopes

Recall from that in each of the 25 hospitals, four wards were selected and randomly assigned to an experimental and control condition as an example of a cluster-randomized study. Thus, experimental condition is a ward-level (i.e., Level 2) explanatory variable. Just as the effects of Level 1 regressors (e.g., nurse experience) can potentially vary across Level 2 units (e.g., wards) or even Level 3 units, it's also possible for the effects of Level 2 regressors to vary across Level 3 units. In the nurse example, we want to estimate the effect of the ward-level experimental training program on nurse stress. It's possible that this training program was more effective in some hospitals than others, suggesting that the treatment effect could vary across hospitals.

But it may also be important to control nurse experience statistically, so we can expand the previous model in which the effect of nurse experience randomly varies across wards to also include the effect of the training program randomly varying across hospitals. (But the results above suggest that we don't need to allow the experience effect to vary across hospitals.) The new model is thus

Level 1: $Y_{ijk} = \beta_{0jk} + \beta_{1jk} X_{ijk\bullet} + \varepsilon_{ijk}$

$stress_{ijk} = \beta_{0jk} + \beta_{1jk} experience_{ijk\bullet} + \varepsilon_{ijk}$

Level 2a: $\beta_{0jk} = \gamma_{00k} + \gamma_{01k} w_{jk} + \delta_{0jk}$

$\beta_{0jk} = \gamma_{00k} + \gamma_{01k} treatment_{jk} + \delta_{0jk}$ (random effect of *treatment* on ward-level intercept)

Level 2b: $\beta_{1jk} = \gamma_{10k} + \delta_{1jk}$ (random ward-level slope of *experience*)

Level 3a: $\gamma_{00k} = \mu_{000} + \varsigma_{00k}$ (random hospital-level intercept)

Level 3b: $\gamma_{10k} = \mu_{100}$ (fixed hospital-level slope of *experience*)

Level 3c: $\gamma_{01k} = \mu_{010} + \varsigma_{01k}$ (random hospital-level slope of *treatment*)

It's also possible to include *treatment* in the Level 2b equation for the ward-level *experience* slopes, which would imply a cross-level, *treatment* by *experience* interaction (perhaps the training program more effectively reduces *stress* among nurses with less experience?), but we will leave this possibility for readers to investigate on their own.

After fitting this model to the data, the fixed-effect estimates of interest are:

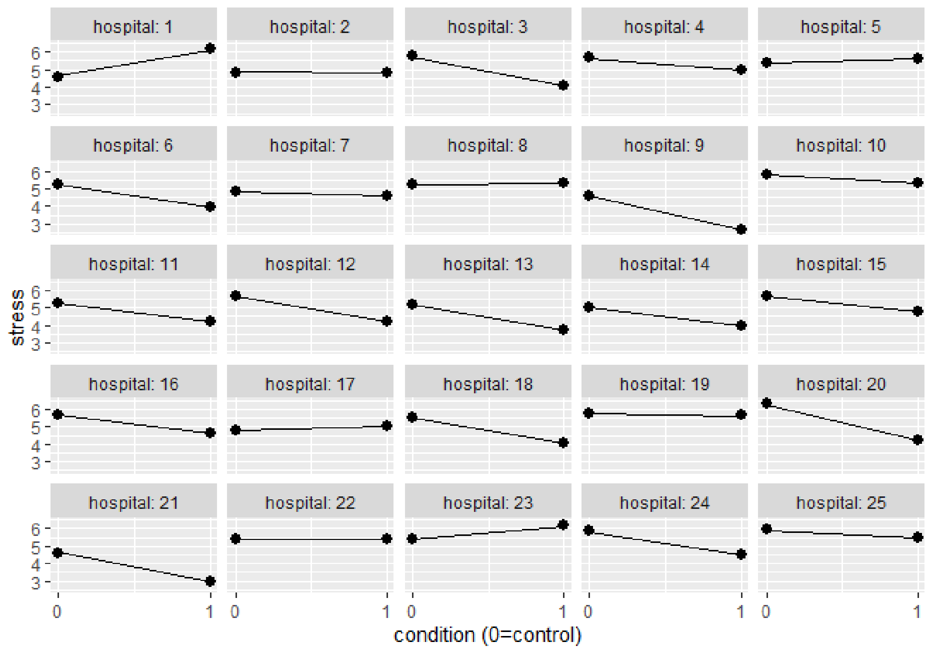
$\hat{\mu}_{000}$ = 5.35 is the predicted level of *stress* for a nurse whose ward is in the control condition and whose experience equals the mean *experience* in the same ward (because *experience* was cluster-mean centered).

$\hat{\mu}_{100}$ = -0.24 is the predicted decrease in *stress* when comparing a nurse with an additional year of *experience* with another nurse in the same ward (because *experience* was cluster-mean centered), holding the effect of the training program constant.

$\hat{\mu}_{010}$ = -0.70 indicates that the mean *stress* of wards assigned to receive the training program is 0.70 units of *stress* lower than the mean *stress* of wards in the control condition, holding nurse experience constant.

But the random-effects output indicates that the effect of the training program varies substantially across hospitals, with SD (ζ_{01k}) = 0.82.

A plot of the within-hospital predicted means by condition shows that the treatment effect does vary quite a bit across hospitals:



Reference

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255-278.